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Method

HARP-FiLM: Global Ionospheric VTEC Forecasting With Solar–Geomagnetic Driver Conditioning and Dual-Baseline Residuals

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Abstract

Accurate 24 hr global ionospheric vertical total electron content (VTEC) forecasting is difficult because diurnal recurrence, solar-cycle variability, and geomagnetic disturbances act on different time scales. We present HARP-FiLM (Horizon-Aware Adaptive Residual Prediction with Feature-wise Linear Modulation), a solar–geomagnetic-driver-conditioned framework that combines multi-scale spatial encoding, Transformer temporal modeling, structured feature modulation, and adaptive dual-baseline residual prediction. Using global ionosphere maps from the Center for Orbit Determination in Europe (CODE) and OMNI2 drivers, HARP-FiLM achieves root-mean-square errors (RMSEs) of 1.356 TECU in 2020 and 4.956 TECU in 2023. Relative to the International Reference Ionosphere 2016 (IRI-2016), RMSE is reduced by 57.7% and 56.3%, while blocked experiments yield reductions of 62.9% and 47.4% in

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reported skill is not primarily produced by direct fitting to temporally interpolated targets. External cross-product evaluation against the Institute for Space–Earth Environmental Research (ISEE) ground-based GNSS-derived TEC product also shows lower errors than IRI-2016 in all four evaluated years, although the margin narrows in 2023. Regional, seasonal, geomagnetic-activity-stratified, and rapid-forcing analyses show the strongest gains under quiet to moderately disturbed conditions, whereas intense and super-storm periods remain challenging. These results support structured driver conditioning and adaptive temporal priors for global VTEC forecasting while highlighting limitations associated with severe disturbances and gridded target products.

Plain Language Summary

The ionosphere affects satellite navigation and radio signals because its electron content changes with local time, season, solar activity, and geomagnetic disturbances. These changes are especially difficult to forecast during strong space-weather events. We developed a global one-day forecasting method that combines recent ionospheric maps with information about solar and geomagnetic conditions. The method also uses recent observations and the ionospheric state at the same time on the previous day as adaptive reference patterns. Tests across years with different solar activity show lower forecast errors than a standard empirical ionosphere model, simple persistence methods, and several representative learning models. A separate comparison with a gridded electron-content product derived from ground-based satellite-navigation observations shows that the improvement is not limited to agreement with the maps used for training. The largest benefits occur under quiet to moderately disturbed conditions, while the strongest geomagnetic storms remain difficult to predict. These results show that structured use of space-weather information can improve global ionospheric forecasting while also identifying where additional storm-time observations and modeling are still needed.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

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The CODE Global Ionosphere Map (GIM) data used as the target VTEC data set in this study are from the CODE final product series for the IGS, provided by the Center for Orbit Determination in Europe (CODE) at the Astronomical Institute, University of Bern (Dach et al., **2026**). The hourly solar-wind and geomagnetic driver data used in this study are available from the NASA OMNIWeb OMNI2 database provided by the NASA Goddard Space Flight Center Space Physics Data Facility via **https://omniweb.gsfc.nasa.gov** (NASA Goddard Space Flight Center, **2026**). Version 1.0.1 of the HARP-FiLM source code used to train, evaluate, and reproduce the forecasting model is preserved in Zenodo via **https://doi.org/10.5281/zenodo.20406934**, licensed under the MIT License, and developed openly at **https://github.com/research1011/HARP-FiLM** (Chen, Yang, & Han, **2026**). The ISEE global GNSS-derived absolute TEC AGRID2 NetCDF products used for the external cross-product evaluation are publicly available from the Institute for Space–Earth Environmental Research (ISEE), Nagoya University, through the ISEE GNSS-TEC database (Shinbori et al., **2020**, **2021**; Zossi et al., **2026**). The AGRID2 data used in this study were downloaded from **https://stdb2.isee.nagoya-u.ac.jp/GPS/shinbori/AGRID2/nc/**.

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